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Estimating long-run relationships from dynamic heterogeneous panels

M. Hashem Pesaran*[, a], Ron Smith[b]

[a]Trinity College, Cambridge CB2 1TQ, UK
[b]Department of Economics, Birkbeck College, London W1P 1PA, UK

Abstract

In panel data four procedures are widely used: pooling, aggregating, averaging group estimates, and cross-section regression. In the static case, if the coefficients differ randomly, all four procedures give unbiased estimates of coefficient means. In the dynamic case, when the coefficients differ across groups, pooling and aggregating give inconsistent and potentially highly misleading estimates of the coefficients, though the cross-section can provide consistent estimates of the long-run effects. The theoretical results on the properties of the four procedures are illustrated by UK labour demand functions for 38 industries over 30 years.

Key words: Dynamic panels; Parameter heterogeneity; Data fields; Sectoral employment equations
JEL classification: C13; C21; C22; C23

1. Introduction

It is now quite common to have panels in which both N (the number of groups) and T (the number of time periods) are quite large. The groups may be firms, industries, regions, or countries (as in the Summers–Heston data set widely used to study growth). Quah (1990) calls such panels 'data fields' to distinguish them from the small T panels typical of micro-econometrics. Surveys

* Corresponding author.

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of the literature on panel data are provided in Hsiao (1986) and Matyas and Sevestre (1992).

Suppose the parameter of interest is the average effect of some exogenous variable on a dependent variable. When T is large enough that it is sensible to run separate regressions for each group, there are four procedures that can be used to estimate this average effect. The first involves estimating separate regressions for each group and averaging the coefficients over groups. We will refer to this as the mean group estimator. The second involves combining the data by imposing common slopes, allowing for fixed or random intercepts, and estimating pooled regressions. This coefficient homogeneity assumption although very widely adopted seems implausible for many applications (e.g., see the discussion in Mairesse and Griliches, 1990). The third involves averaging the data over groups and estimating aggregate time-series regressions. The properties of time-series regressions on group averages are examined in more detail in Pesaran, Pierse, and Kumar (1989) and Lee, Pesaran, and Pierse (1990). The fourth involves averaging the data over time and estimating cross-section regressions on group means. Much of the recent work on endogenous growth has used this approach (e.g., Barro, 1991). All four procedures will provide an estimate of the average coefficient; the difference is that in the case of the first estimator the averaging is explicit, while in the others it is implicit.

In the static case, where the regressors are strictly exogenous and the coefficients differ randomly and are distributed independently of the regressors across groups, all four procedures provide consistent (and unbiased) estimates of the coefficient means (e.g., Zellner, 1969). It often seems to be assumed that a similar result holds for dynamic models, i.e., all four procedures give consistent estimators.[1] We show that this is not the case. The group mean estimator obtained by averaging the coefficients for each group is consistent for large N and T, and thus provides a standard of comparison. However, the pooled and aggregate estimators are not consistent in dynamic models, even for large N and T, and the biases can be very substantial. The problem arises because when the regressors are serially correlated, incorrectly ignoring coefficient heterogeneity induces serial correlation in the disturbance, which generates inconsistent estimates in models with lagged dependent variables, even as $T \to \infty$. Thus this inconsistency is quite distinct from that suffered by the fixed effect estimator in small T panels as $N \to \infty$ (e.g., Nickell, 1981), and the solutions proposed in that literature do not solve this problem. Interestingly, the cross-section regressions when based on long time averages do produce consistent estimates of the average long-run coefficients, confirming a widely believed piece of econometric

[1] Certainly Zellner believed this to be the case. Zellner (1969, footnote 5) says: 'if X contains lagged values of y, (13) $\{E\hat{\beta} = \bar{\beta}\}$ would have to be reformulated to read plim $\hat{\beta} = \bar{\beta}$, a point made by H. Theil in conversation with the author' (term in { } added).

folk wisdom (e.g., Baltagi and Griffin, 1984). However, it should be emphasised that the familiar practice of running cross-section regressions based on a single or a few years of observations is not likely to yield unbiased or consistent estimates.

Section 2 examines the properties of the four procedures when the regressors are stationary and the model includes a lagged dependent variable. Section 3 examines their properties when the regressors are integrated and each group relationship cointegrates. Section 4 applies the various procedures to the data used in Pesaran, Pierse, and Kumar (1989) and Lee, Pesaran, and Pierse (1990), henceforth referred to as PPK and LPP, respectively, who estimated labour demand functions for 38 industries in the U.K. over 29 years. The application confirms that the biases in the pooled and aggregate estimates can be very serious. Section 5 contains some concluding remarks.

2. Dynamic models of heterogeneous panels

We assume that the data are generated by a set of relationships which have coefficients that are constant over time but differ randomly across groups, and that the distribution of the coefficients is independent of the regressors.[2] The parameters of interest are the averages, over groups, of the coefficients. Consider the following simple heterogeneous dynamic model:[3]

$$y_{it} = \lambda_i y_{i,t-1} + \beta_i' \boldsymbol{x}_{it} + \varepsilon_{it}, \qquad i = 1,2,\ldots,N, \quad t = 1,2,\ldots,T, \tag{2.1}$$

with its coefficients λ_i and β_i varying across groups according to the following random coefficient model:

$$\mathrm{H_a}: \quad \lambda_i = \lambda + \eta_{1i}, \quad \beta_i = \beta + \eta_{2i}, \tag{2.2}$$

where η_{1i} and η_{2i} are assumed to have zero means and constant covariances. We also assume that the higher-order moments of η_{1i} and η_{2i} and their cross-moments exist and are finite. This is the standard formulation of the Random Coefficients Model (RCM), and introduces parameter heterogeneity through the 'short-run' coefficients, β_i and λ_i. An alternative characterization would be to let the long-run effects, $\theta_i = \beta_i/(1-\lambda_i)$, and the mean lags, $\psi_i = \lambda_i/(1-\lambda_i)$, vary randomly across groups:

$$\mathrm{H_b}: \quad \psi_i = \psi + \xi_{1i}, \quad \theta_i = \theta + \xi_{2i}, \tag{2.3}$$

[2] The assumption of randomness is made for convenience. The main results, in particular the inconsistency of the pooled estimator, will hold in the case where the coefficients are fixed but differ across groups.

[3] For expositional simplicity we confine our attention to the simple specification (2.1), but the main results of the paper also apply to more complicated dynamic models.

where it is now assumed that ξ_{1i} and ξ_{2i} have zero means and constant covariances. The 'average' long-run effect of $\boldsymbol{x}$ on y can now be defined either in terms of the 'average' of the short-run coefficients, $\bar{\beta}/(1-\bar{\lambda})$, or the average of the long-run coefficients, $\bar{\theta}$, where

$$\bar{\beta} = N^{-1}\sum_{i=1}^{N}\beta_i,\quad \bar{\lambda} = N^{-1}\sum_{i=1}^{N}\lambda_i,\quad \bar{\theta} = N^{-1}\sum_{i=1}^{N}\theta_i.$$

As $N \to \infty$, these two estimates converge to $\mathrm{E}(\beta_i)/(1-\mathrm{E}(\lambda_i))$ and $\mathrm{E}(\beta_i/(1-\lambda_i)) = \mathrm{E}(\theta_i)$, respectively, which in general differ from one another, unless λ_i and θ_i are independently distributed, for all i. It turns out that $\bar{\theta}$ [or its limit as $N \to \infty$, namely $\mathrm{E}(\theta_i)$] is the appropriate measure of the average long-run effect of $\boldsymbol{x}$ on y, which is also the probability limit of the cross-section estimates obtained in Section 2.3.

Throughout the paper, unless specifically stated otherwise, we shall make use of the following assumptions:

Assumption 1. $\boldsymbol{x}_{it}$ and ε_{is} are independently distributed for all t and s (i.e., $\boldsymbol{x}_{it}$'s are strictly exogenous), and both set of variables are independently distributed of η_{1i} and η_{2i}, under $\mathrm{H_a}$, and of ξ_{i1} and ξ_{i2}, under $\mathrm{H_b}$. The disturbances ε_{it} have zero means and variances, σ_i^2, that are constant over time.

Assumption 2. $\boldsymbol{x}_{it}$'s are covariance-stationary and mean square ergodic processes with means $\boldsymbol{\mu}_i$.[4]

Assumption 3. The support of the random coefficients λ_i lie in the stable range $(-1, 1)$. Further, all the cross-moments of λ_i and β_i exist and are finite, and in particular $\mathrm{E}(\beta_i/(1-\lambda_i))$ and $\mathrm{E}(1/(1-\lambda_i))$ exist and are finite.

Assumption 4. The cross-product matrices $N^{-1}\sum_{i=1}^{N}\boldsymbol{\mu}_i\boldsymbol{\mu}_i'$ are nonsingular for a finite N, and converge to a finite nonsingular matrix as $N \to \infty$.

Under the above assumptions and when data fields (namely panels with N and T large) are available, separate regressions can be run for each group and

[4] The necessary and sufficient condition for a covariance stationary process $\{x_t\}$ to be mean square ergodic is given by

$$\lim_{T\to\infty}\left\{T^{-1}\sum_{s=0}^{T}\gamma(s)\right\} = 0,$$

where $\gamma(s)$ is the autocovariance function of the process. This condition ensures that time averages such as $\bar{\boldsymbol{x}}_i$ converge to the means of $\boldsymbol{x}_{it}$, namely μ_i in the mean square error sense as $T \to \infty$. See, for example, Karlin and Taylor (1975, § 9.5).

the estimates obtained can be averaged. Both the unweighted average and the Generalised Least Squares weighted average proposed by Swamy (1971) will yield consistent estimates of λ and β and of θ and ψ under both H_a and H_b, for both N and T large. Of course, for small T, the estimates β_i and λ_i will inevitably be biased because of the presence of the lagged dependent variable and this will give an inconsistent estimate of their mean even for large N. However, in empirical applications estimating N separate regressions is rather rare, it is much more common to pool or average the data. Given that they are very widely used, it is important to examine the properties of the other three estimation procedures discussed in Section 1 – pooled, aggregate time-series, and cross-section – to the dynamic heterogeneous panel model (2.1). For exposition, the pooled and aggregate time-series estimator will be examined just under H_a. The results under H_b are qualitatively similar, and the estimates remain inconsistent even when $T \to \infty$. However, the cross-section estimator will be examined under both random coefficient hypotheses H_a and H_b as far as the estimation of average long-run effects are concerned, the (asymptotic) results do not depend on the particular specification and the random coefficient model.

2.1. Pooled estimators

Suppose the micro relations are specified by (2.1) explicitly including an intercept term, with random parameters defined according to H_a in (2.2). The pooled regression is given by

$$y_{it} = \alpha_i + \lambda y_{i,t-1} + \beta' \boldsymbol{x}_{it} + v_{it}, \tag{2.4}$$

$$i = 1, 2, \ldots, N, \quad t = 1, 2, \ldots, T,$$

$$v_{it} = \varepsilon_{it} + \eta_{1i} y_{i,t-1} + \eta'_{2i} \boldsymbol{x}_{it}. \tag{2.5}$$

Different group-specific fixed or random effects can be included in the pooled regressions through the intercept term α_i. It is now easily seen that under Assumptions 1–3 $y_{i,t-1}$ and $\boldsymbol{x}_{it}$ are correlated with v_{it}, thus rendering the pooled estimators inconsistent. More specifically, in the case where $\boldsymbol{x}_{it}$ is a scalar with zero mean and autocovariance function $\gamma_i(s)$, we have

$$\mathrm{E}(\boldsymbol{x}_{it} v_{it}) = \sum_{j=0}^{\infty} \mathrm{E}(\eta_{1i} \beta_i \lambda_i^j) \gamma_i(|j+1|)$$

and

$$\mathrm{E}(y_{i,t-1} v_{it}) = \sum_{s=0}^{\infty} \sum_{r=0}^{\infty} \mathrm{E}(\eta_{1i} \beta_i^2 \lambda_i^{r+s}) \gamma_i(|r-s|) + \sigma_i^2 \sum_{s=1}^{\infty} \mathrm{E}(\eta_{1i} \lambda_i^{2s})$$
$$+ \sum_{s=0}^{\infty} \mathrm{E}(\eta_{2i} \beta_i \lambda_i^s) \gamma_i(|s+1|).$$

Even if $\boldsymbol{x}_{it}$'s are serially uncorrelated, $\mathrm{E}(y_{i,t-1}\, v_{it})$ does not vanish because of the second term. In addition, the problem is unlikely to be amenable to the usual solutions. Standard corrections for serial correlation are unlikely to work because of the complexity of the process generating v_{it}. Nor is estimation by the instrumental variables (IV) method likely to be successful either. Given the structure of the composite disturbances v_{it}, all variables that are correlated with $y_{i,t-1}$ or $\boldsymbol{x}_{it}$ will also be correlated with v_{it}. In general, only variables that are uncorrelated with lagged values of ε_{it} and $\boldsymbol{x}_{it}$ have a zero correlation with v_{it}. But such variables, assuming they exist, fail to yield a valid set of instruments, since they will also be uncorrelated with the regressors. The fact that the average effects in a dynamic model where the slope coefficients differ over groups cannot be consistently estimated from a pooled regression even when N and T are large is not a particular problem since they can be estimated directly by the mean group procedure. Of course, when T is small the bias of the pooled estimator is likely to be a serious problem.

Since the pooled estimators are widely used, it would be useful to have some idea of the size of the likely biases and what they depend on. In the case where the slope coefficients are homogeneous, it is well-known that the 'within', or the fixed-effect estimators, and the random-effect estimators (namely the GLS estimators of λ and β when α_i's are treated as random) are asymptotically equivalent for both N and T large (see, for example, Chapter 4 in Matyas and Sevestre, 1992). Consequently, here we focus on the fixed-effect estimators, and expect to obtain similar results when the random-effect estimators are considered. Also to simplify the exposition we analyze the case where the model contains only one exogenous variable ($k = 1$), generated according to the following simple stationary autoregressive process:

$$x_{it} = \mu_i(1-\rho) + \rho\, x_{i\,t-1} + u_{it}, \tag{2.6}$$

where μ_i is the (unconditional) mean of x_{it}, $|\rho| < 1$, and for each i, $u_{it} \sim \mathrm{iid}(0, \tau_i^2)$. The variations in σ_i^2 and τ_i^2 across groups are also supposed to be independent of λ_i and β_i. We shall denote the means of σ_i^2 and τ_i^2 by σ^2 and τ^2, respectively.[5]

The probability limits of the fixed-effect estimators, which we denote by $\hat{\lambda}$ and $\hat{\beta}$, are derived in Appendix A. It is shown that the fixed-effect estimators are generally inconsistent (for large N and T), and yield an inconsistent estimator of the 'average' long run effects of $\boldsymbol{x}$ on y. Furthermore, this inconsistency does not disappear even if variations in β_i across groups is the only source of slope heterogeneity in the model. For this relatively simple case (with $\lambda_i = \lambda$, and

[5] When σ_i^2 and τ_i^2 are nonstochastic but variable across groups, σ^2 and τ^2 can be viewed as the limits of their respective averages.

hence $\eta_{1i} = 0$ for all i) we have[6]

$$\operatorname*{plim}_{N\to\infty,T\to\infty}(\hat{\lambda}) = \lambda + \frac{\rho(1-\lambda\rho)(1-\lambda^2)\omega_{22}}{\Psi_1}, \tag{2.7}$$

$$\operatorname*{plim}_{N\to\infty,T\to\infty}(\hat{\beta}) = \beta - \frac{\beta\rho^2(1-\lambda^2)\omega_{22}}{\Psi_1}, \tag{2.8}$$

where

$$\Psi_1 = (\sigma^2/\tau^2)(1-\rho^2)(1-\lambda\rho)^2 + (1-\lambda^2\rho^2)\omega_{22} + (1-\rho^2)\beta, \tag{2.9}$$

and ω_{22} is the variance of β_i. Since $|\lambda| < 1$ and $|\rho| < 1$, then $\Psi_1 > 0$. The large sample bias of $\hat{\lambda}$ will be positive when $\rho > 0$, and *vice versa*, while $\operatorname{plim}(\hat{\beta}) < \beta$ for all parameter values. The size of the asymptotic bias of the fixed-effect estimators depends on ρ, λ, β, ω_{22}, and σ^2/τ^2. As to be expected, the larger the degree of parameter heterogeneity (i.e., the larger ω_{22}), the greater the bias. For $\omega_{22} > 0$, the asymptotic biases in $\hat{\lambda}$ and $\hat{\beta}$ disappears only if $\rho = 0$.[7] Turning to the fixed-effect estimator of the 'average' long-run impact of x on y, namely $\hat{\theta} = \hat{\beta}/(1-\hat{\lambda})$, using (2.7) and (2.8) we have

$$\operatorname*{plim}_{N\to\infty,T\to\infty}(\hat{\theta}) = \frac{\theta}{1-\rho\Psi_2}, \tag{2.10}$$

where

$$\Psi_2 = \frac{(1+\lambda)\omega_{22}}{(\sigma^2/\tau^2)(1+\rho)(1-\lambda\rho)^2 + (1+\rho)(\beta^2+\omega_{22})} > 0\,. \tag{2.11}$$

Therefore, the fixed-effect estimator of the long-run effect, θ, is also asymptotically biased, and overestimates the long-run effect when $\rho > 0$, and underestimates it when $\rho < 0$. The bias disappears either when $\rho = 0$ or $\omega_{22} = 0$.[8]

As ρ approaches unity from below ($|\rho| < 1$), we have the following important results:

$$\operatorname*{plim}_{\rho\to 1}(\hat{\lambda}) = 1 \quad \text{and} \quad \operatorname*{plim}_{\rho\to 1}(\hat{\beta}) = 0,$$

which hold irrespective of the true value of λ![9] Robertson and Symons (1992) also obtain a similar result by considering pooled OLS estimates in the context

[6] Derivations are given in Appendix A. The probability limits are obtained by first letting $T \to \infty$, and then $N \to \infty$.

[7] This result crucially depends on the assumed homogeneity of λ_i and ρ_i across groups.

[8] The consistency of $\hat{\theta}$ for θ when $\rho = 0$ does not carry-over to the more general dynamic model where (2.1) also contains lagged values of $\boldsymbol{x}_{it}$.

[9] It is important to note that these results hold only for values of ρ approaching unity from below (i.e., when $\boldsymbol{x}_{it}$'s are near integrated) and are not valid for $\rho = 1$.

of a dynamic heterogeneous panel when the true value of λ is zero. Our results show that Robertson and Symons' finding is more general and holds for *all* values of $|\lambda| < 1$. The tendency for the fixed-effect estimators to underestimate the average short-run effects and overestimate the average long-run effects can be clearly seen in the application considered in Section 4, even though the sectoral employment demand equations are specified to contain higher-order lags of the dependent and the explanatory variables. Finally, it is worth noting that for $\rho \neq 0$ and $\omega_{22} \neq 0$ the asymptotic biases of $\hat{\lambda}$ and $\hat{\beta}$ vanish only if the true value of λ approaches its upper limit of unity from below. Using (2.7) and (2.8), it is clear that when $\lambda \to 1$, $\text{plim}(\hat{\lambda})$ and $\text{plim}(\hat{\beta})$ tend to λ and β, respectively. Once again this result is not necessarily valid for $\lambda = 1$.

2.2. *The aggregate time-series estimator*

As with the pooled regression, if the $\boldsymbol{x}_{it}$ are serially correlated, then the error in the aggregate equation will also be serially correlated producing inconsistent estimates if a lagged dependent variable is included in the model. Consider the simple dynamic RCM defined by (2.1) and assume that the short-run coefficients λ_i and β_i are random (i.e., H_a holds). The case where the long-run coefficients are assumed to be random yields the same qualitative results and will not be dealt with here separately.

Aggregating (2.1) over the micro-units and utilizing (2.2) we obtain

$$\bar{y}_t = \lambda \bar{y}_{t-1} + \beta' \bar{\boldsymbol{x}}_t + \bar{v}_t, \tag{2.12}$$

where $\bar{y}_t$ and $\bar{\boldsymbol{x}}_t$ are sample means of y_{it} and $\boldsymbol{x}_{it}$ across i, and

$$\bar{v}_t = \bar{\varepsilon}_t + N^{-1} \sum_{i=1}^{N} (\eta_{1i} y_{i,t-1} + \eta'_{2i} \boldsymbol{x}_{it}).$$

The macro disturbances, $\bar{v}_t$, are now correlated with the regressors in the aggregate equation and the OLS estimators based on (2.12) will not be consistent even if both N and T are allowed to increase without bounds. (See Appendix B.) As with the pooled regression, the serial correlation pattern in $\bar{v}_t$ is also sufficiently complex that standard procedures for dealing with it are unlikely to work and estimation of (2.12) by the instrumental variables (IV) method is not likely to be successful either. For example, lagged $\bar{x}_t$'s cannot be used as instruments, since

$$\text{E}(\bar{\boldsymbol{x}}_{t-r} \bar{v}_t) = N^{-1} \sum_{i=1}^{N} \sum_{s=0}^{\infty} \text{E}(\eta_{1i} \lambda_i^s \beta'_i) \, \text{E}(\boldsymbol{x}_{i,t-s-1} \bar{\boldsymbol{x}}_{t-r}),$$

which is nonzero for all values of $r > 0$. In general, only variables that are uncorrelated with lagged values of ε_{it} and $\boldsymbol{x}_{it}$ have a zero correlation with $\bar{v}_t$. But such variables, assuming they exist, fail to yield a valid set of instruments, since they will also be uncorrelated with the regressors of (2.12), namely $\bar{y}_{t-1}$ and $\bar{x}_t$. Therefore, Zellner's (1969) ingenious solution of the aggregation problem in the context of linear models does not, unfortunately, extend to random coefficient micro-relations with lagged endogenous variables.

The difficulty in estimating λ and β from the above aggregate time-series specification partly stems from the fact that (2.12) is not an 'optimal' aggregate predictor of $\bar{y}_t$, given the available aggregate information set, $\Omega_t = (\bar{\boldsymbol{x}}_t, \bar{\boldsymbol{x}}_{t-1}, \ldots; \bar{y}_{t-1}, \bar{y}_{t-2}, \ldots)$. As one of us has shown elsewhere (Pesaran, 1995), the optimal aggregate time-series specification associated with the heterogeneous dynamic model (2.1) is given by

$$\bar{y}_t = \sum_{j=0}^{\infty} \boldsymbol{a}'_j \bar{\boldsymbol{x}}_{t-j} + \bar{\xi}_t, \tag{2.13}$$

where $\bar{\xi}_t$'s have zero means and are independently distributed of the current and lagged values of $\bar{\boldsymbol{x}}_t$, and

$$\boldsymbol{a}_j = \mathrm{E}(\beta_i \lambda_i^j) \quad \text{for} \quad j = 0, 1, 2, \ldots . \tag{2.14}$$

In general, $\bar{\xi}_t$'s will have a complicated pattern of serial correlation. Comparing (2.13) with (2.12) it is now clear that (2.12) is misspecified and cannot be used to obtain consistent estimators of λ and β. A special case of (2.13) is also derived by Lewbel (1994), under the assumption that β_i and λ_i are independently distributed [i.e., $\operatorname{cov}(\eta_{1i}, \eta_{2i}) = 0$, $\forall i$]. However, to obtain the 'average' long-run effect of $\boldsymbol{x}_{it}$ on y_{it}, one does not need to make the restrictive assumption that λ_i and β_i are independently distributed, and using (2.14), it is easily seen that

$$\sum_{j=0}^{\infty} \boldsymbol{a}_j = \mathrm{E}\left(\sum_{j=0}^{\infty} \beta_i \lambda_i^j\right) = \mathrm{E}\left(\frac{\beta_i}{1-\lambda_i}\right) = \mathrm{E}(\theta_i) = \theta. \tag{2.15}$$

Hence, to estimate θ consistently from an aggregate times specification one needs to consider an infinite distributed lag specification between $\bar{y}_t$ and $\bar{\boldsymbol{x}}_t$, allowing for the possibility of a high degree of residual serial correlation.

2.3. The cross-section estimator

Aggregating (2.1) over time, we get

$$\tilde{y}_i = \beta'_i \tilde{\boldsymbol{x}}_i + \lambda_i \tilde{y}_{i,-1} + \tilde{\varepsilon}_i, \tag{2.16}$$

where

$$\tilde{y}_{i,-1} = T^{-1} \sum_{t=1}^{T} y_{i,t-1}, \qquad \tilde{\boldsymbol{x}}_i = T^{-1} \sum_{t=1}^{T} x_{it}, \qquad \tilde{\varepsilon}_i = T^{-1} \sum_{t=1}^{T} \varepsilon_{it}.$$

Under $\mathrm{H_a}$, defined by (2.2), (2.16) becomes

$$\tilde{y}_i = \beta' \tilde{\boldsymbol{x}}_i + \lambda \tilde{y}_{i,-1} + \tilde{v}_i, \tag{2.17}$$

where

$$\tilde{v}_i = \eta_{1i}\tilde{y}_{i,-1} + \eta'_{2i}\tilde{\boldsymbol{x}}_i + \tilde{\varepsilon}_i. \tag{2.18}$$

The regression defined by (2.17) is the 'between' regression for the dynamic model, and will produce inconsistent estimates of β and λ as $N, T \to \infty$, even if the intercept parameter in (2.1) is the *only* parameter that varies across groups. This is due to the fact that $\tilde{v}_i$ is correlated with $\tilde{y}_{i,-1}$ even for large T.

A more promising route is first to rewrite (2.16) as

$$\tilde{y}_i = \beta'_i\tilde{\boldsymbol{x}}_i + \lambda_i(\tilde{y}_i - \Delta_T y_i) + \tilde{\varepsilon}_i, \tag{2.19}$$

where the growth term $\Delta_T y_i = (y_{iT} - y_{i0})/T$ captures the end effects. Then solve for $\tilde{y}_i$ to obtain

$$\begin{aligned}\tilde{y}_i &= \frac{\beta'_i}{1-\lambda_i}\tilde{\boldsymbol{x}}_i - \frac{\lambda_i}{1-\lambda_i}\Delta_T y_i + \frac{1}{1-\lambda_i}\tilde{\varepsilon}_i \\ &= \theta'_i\tilde{\boldsymbol{x}}_i - \psi_i\Delta_T y_i + \frac{1}{1-\lambda_i}\tilde{\varepsilon}_i.\end{aligned}$$

Thus the coefficients $\tilde{\boldsymbol{x}}_i$ in this cross-section regression are in fact the appropriate long-run averages. Given the structure of this equation, it is natural to assume that the randomness is introduced through the long-run coefficients. Thus, under $\mathrm{H_b}$, the cross-section regression is

$$\tilde{y}_i = \theta'\tilde{\boldsymbol{x}}_i - \psi\Delta_T y_i + \tilde{u}_i, \tag{2.20}$$

where

$$\tilde{u}_i = \frac{1}{1-\lambda_i}\tilde{\varepsilon}_i + \xi'_{2i}\tilde{\boldsymbol{x}}_i - \xi_{1i}\Delta_T y_i, \tag{2.21}$$

and the cross-section estimates of the 'average' long-run coefficients are given by

$$\hat{\theta}^c = \left(\sum_{i=1}^{N}\tilde{\boldsymbol{x}}_i\tilde{\boldsymbol{x}}'_i\right)^{-1}\left(\sum_{i=1}^{N}\tilde{\boldsymbol{x}}_i\tilde{y}_i\right). \tag{2.22}$$

It may appear that the usual cross-sections, estimated purely from levels, are misspecified to the extent that they omit the growth terms $\Delta_T y_i$. However, asymptotically the growth terms such as $\Delta_T y_i$ (and $\Delta_T \boldsymbol{x}_i$ had $\boldsymbol{x}_{i,t-1}$ appeared in the regression) are uncorrelated with the level terms, so omitting them is irrelevant to the consistency of the average long-run effects. This suggests that cross-section estimates of the average long-run effect, θ, will be robust to dynamic misspecification of the underlying micro-model. This robustness is clearly an important feature of cross-section estimates of θ, not enjoyed by pooled or aggregate time-series estimates.

Consider now the asymptotic bias of the cross-section estimate, $\hat{\theta}^c$, for large T. First, note that since $\psi_i = \lambda_i/(1-\lambda_i)$, then $1 + \psi_i = 1/(1-\lambda_i)$, and using (2.21), we have

$$\tilde{u}_i = (1+\psi_i)\tilde{\varepsilon}_i + \tilde{\boldsymbol{x}}_i'\xi_{2i} - \xi_{1i}\Delta_T y_i. \tag{2.23}$$

Also, under Assumption 3 and assuming that the processes generating y_{it} have started a long time ago, we have[10]

$$y_{it} = \sum_{j=0}^{\infty} \boldsymbol{x}_{i,t-j}'\beta_i\lambda_i^j + \sum_{j=0}^{\infty}\lambda_i^j\varepsilon_{i,t-j}, \tag{2.24}$$

and, therefore,

$$\begin{aligned}\Delta_T y_i &= T^{-1}(y_{iT}-y_{i0})T^{-1}\sum_{t=1}^{T}\Delta y_{it}\\ &= \sum_{j=0}^{\infty}(\tilde{\boldsymbol{x}}_{i,-j}-\tilde{\boldsymbol{x}}_{i,-j-1})'\beta_i\lambda_i^j + \sum_{j=0}^{\infty}\lambda_i^j(\tilde{\varepsilon}_{i,-j}-\tilde{\varepsilon}_{i,-j-1}).\end{aligned} \tag{2.25}$$

Substituting (2.20) for $\tilde{y}_i$ in (2.22) and making use of (2.23) and (2.25), we have

$$\begin{aligned}\hat{\theta}^c - \theta &= \boldsymbol{H}_N\sum_{i=1}^{N}(\tilde{\boldsymbol{x}}_i\tilde{\boldsymbol{x}}_i')\xi_{2i} + \boldsymbol{H}_N\sum_{i=1}^{N}(1+\psi_i)\tilde{\boldsymbol{x}}_i\tilde{\varepsilon}_i - \boldsymbol{H}_N\\ &= \sum_{i=1}^{N}\sum_{j=0}^{\infty}\tilde{\boldsymbol{x}}_i(\tilde{\boldsymbol{x}}_{i,-j}-\tilde{\boldsymbol{x}}_{i,-j-1})'\psi_i\beta_i\lambda_i^j\\ &\quad - \boldsymbol{H}_N\sum_{i=1}^{N}\sum_{j=0}^{\infty}\psi_i\lambda_i^j(\tilde{\varepsilon}_{i,-j}-\tilde{\varepsilon}_{i,-j-1}),\end{aligned}$$

where

$$\boldsymbol{H}_N = \left(\sum_{i=1}^{N}\tilde{\boldsymbol{x}}_i\tilde{\boldsymbol{x}}_i'\right)^{-1}.$$

Denote the set of observations on $\boldsymbol{x}_{it}$'s by $\boldsymbol{x} = (\boldsymbol{x}_{11}, \boldsymbol{x}_{12}, \ldots, \boldsymbol{x}_{1T}; \ldots; \boldsymbol{x}_{N1}, \boldsymbol{x}_{N2}, \ldots, \boldsymbol{x}_{NT})$, and take conditional expectations under Assumption 1 and hypothesis H_b, then

$$\mathrm{E}(\hat{\theta}^c|\boldsymbol{x}) - \theta = -\boldsymbol{H}_N\sum_{j=0}^{\infty}\sum_{i=1}^{N}\tilde{\boldsymbol{x}}_i(\tilde{\boldsymbol{x}}_{i,-j}-\tilde{\boldsymbol{x}}_{i,-j-1})'\,\mathrm{E}(\psi_i\beta_i\lambda_i^j).$$

Under our assumptions the moments $\mathrm{E}(\psi_i\beta_i\lambda_i^j) = \mathrm{E}\{\theta_i[\psi_i/(1+\psi_i)]^{j+1}\}$ exists and are finite for all j. It is also clear that, in general, they are nonzero. It therefore follows that for a *finite* T, the cross-section estimator, $\hat{\theta}^c$, yields

[10] The results are qualitatively unchanged if it is assumed that the initial values y_{i0} are fixed (nonstochastic).

a biased estimator of the mean of the long-run coefficients (i.e., $\theta = \mathrm{E}(\beta_i/(1-\lambda_i))$, and this bias does not vanish for large N. However, for large T (even when N is finite) under the assumption that $\boldsymbol{x}_{it}$'s are covariance-stationary, we have[11]

$$\tilde{\boldsymbol{x}}_i(\tilde{\boldsymbol{x}}_{i,-j} - \tilde{\boldsymbol{x}}_{i,-j-1})' = \mathrm{O}_p(T^{-1}) \quad \text{for} \quad j = 0, 1, 2, \ldots,$$

and the bias will be of order of T^{-1}. Thus for T and N large, the cross-section will provide a consistent estimate of the mean of the long-run coefficients.

2.4. Short-run coefficients and randomness

Consider now the case where the randomness is introduced through the short-run coefficients (i.e., under $\mathrm{H_a}$). Using (2.24), we have the following time averages of y_{it}'s in terms of the time averages of x_{it} and ε_{it}:

$$\bar{y}_i = \sum_{j=0}^{\infty} \tilde{\boldsymbol{x}}_{i,-j}\beta_i\lambda_i^j + \sum_{j=0}^{\infty} \lambda_i^j \tilde{\varepsilon}_{i,-j}, \qquad i = 1, 2, \ldots, N.$$

Substituting this result directly in (2.22) now yields

$$\hat{\theta}^c = \boldsymbol{H}_N^{-1}\left\{ \sum_{j=0}^{\infty}\sum_{i=1}^{N} \tilde{\boldsymbol{x}}_i\tilde{\boldsymbol{x}}'_{i,-j}\beta_i\lambda_i^j + \sum_{j=0}^{\infty}\sum_{i=1}^{N} (\tilde{\boldsymbol{x}}_i\tilde{\varepsilon}_{i,-j})\lambda_i^j \right\}.$$

Taking conditional expectations with respect to $\boldsymbol{x}$ (defined above), and noting that under Assumption 1 the distribution of β_i, λ_i, and ε_{it} are distributed independently of $\boldsymbol{x}_{it}$'s, and under $\mathrm{H_a}$, ε_{it} and λ_i are also distributed independently, we have[12]

$$\mathrm{E_a}(\hat{\theta}^c \,|\, \boldsymbol{x}) = \boldsymbol{H}_N^{-1} \sum_{j=0}^{\infty}\sum_{i=1}^{N} \tilde{\boldsymbol{x}}_i\tilde{\boldsymbol{x}}'_{i,-j}\mathrm{E}(\beta_i\lambda_i^j).$$

But $\mathrm{E}(\beta_i\lambda_i^j) = \boldsymbol{a}_j$ is the same across groups, and hence

$$\mathrm{E_a}(\hat{\theta}^c \,|\, \boldsymbol{x}) = \sum_{j=0}^{\infty} \boldsymbol{H}_N^{-1}\left(\sum_{i=1}^{N} \tilde{\boldsymbol{x}}_i\tilde{\boldsymbol{x}}'_{i,-j}\right)\boldsymbol{a}_j. \tag{2.26}$$

Now for large T and under the assumption that $\boldsymbol{x}_{it}$'s are covariance-stationary, we have (Appendix C)

$$\tilde{\boldsymbol{x}}_i\tilde{\boldsymbol{x}}'_{i,-j} = \underset{\sim}{\mu}_i\underset{\sim}{\mu}'_i + \mathrm{O}_p(T^{-1}), \qquad j = 0, 1, 2, \ldots .$$

[11] A proof in the simple case where $\boldsymbol{x}_{it}$ is a scalar (i.e., when $k = 1$) is given in Appendix C.

[12] Note that $\mathrm{E}(\tilde{\boldsymbol{x}}_i\tilde{\varepsilon}_{i,-j}\lambda_i^j|\boldsymbol{x}) = \tilde{\boldsymbol{x}}_i\mathrm{E}(\tilde{\varepsilon}_{i,-j}\lambda_i^j) = \tilde{\boldsymbol{x}}_i\mathrm{E}(\tilde{\varepsilon}_{i,-j})\mathrm{E}(\lambda_i^j) = 0$, since $\mathrm{E}(\varepsilon_{it}) = 0$ for all i and t.

Therefore,

$$\boldsymbol{H}_N^{-1}\left(\sum_{i=1}^{N} \tilde{\boldsymbol{x}}_i \tilde{\boldsymbol{x}}_{i,-j}'\right) = \left(\sum_{i=1}^{N} \tilde{\boldsymbol{x}}_i \tilde{\boldsymbol{x}}_i'\right)^{-1} \left(\sum_{i=1}^{N} \tilde{\boldsymbol{x}}_i \tilde{\boldsymbol{x}}_{i,-j}'\right)$$
$$= \left\{\sum_{i=1}^{N} \underline{\mu}_i \underline{\mu}_i' + \mathrm{O}_p(T^{-1})\right\}^{-1} \left\{\sum_{i=1}^{N} \underline{\mu}_i \underline{\mu}_i' + \mathrm{O}_p(T^{-1})\right\},$$

and as $T \to \infty$,

$$\operatorname*{plim}_{T \to \infty} \boldsymbol{H}_N^{-1}\left(\sum_{i=1}^{N} \tilde{\boldsymbol{x}}_i \tilde{\boldsymbol{x}}_{i,-j}'\right) = \boldsymbol{I},$$

where $\boldsymbol{I}$ is an identity matrix of the same dimension as $\boldsymbol{x}_i$. Using this result in (2.26), we have [recall that $\boldsymbol{a}_j = \mathrm{E}(\beta_i \lambda_i^j)$]:

$$\operatorname*{plim}_{T \to \infty} \mathrm{E}(\hat{\theta}^c \mid \boldsymbol{x}) = \sum_{j=0}^{\infty} \boldsymbol{a}_j = \mathrm{E}\left(\sum_{j=0}^{\infty} \beta_i \lambda_i^j\right) = \mathrm{E}\left(\frac{\beta_i}{1-\lambda_i}\right) = \theta.$$

Consequently,

$$\operatorname*{plim}_{T \to \infty} \mathrm{E}(\hat{\theta}^c) = \theta,$$

which establishes that for large T cross-section estimator, $\hat{\theta}^c$, is also an unbiased estimator of the average of the long-run coefficients under $\mathrm{H_a}$. Thus, the asymptotic unbiasedness of $\hat{\theta}^c$ does not depend on the particular specification of the random coefficient model.

3. Integrated variables

Suppose that the $\boldsymbol{x}_{it}$ are $I(1)$ and there is a single cointegrating relationship between y_{it} and $\boldsymbol{x}_{it}$ for each group, with the parameters differing randomly between groups. Then a simple static cointegrating regression of y_{it} on $\boldsymbol{x}_{it}$ will yield superconsistent estimates of θ_i, the cointegrating vector. These could then be averaged over groups. However, the pooled and aggregate time-series estimators will not provide consistent estimators of the mean effect. Suppose each micro-relationship cointegrates with different parameters. The pooled regression by imposing a common parameter on all the micro-relations generates a residual which has an $I(0)$ component, the residual from the cointegrating micro-relationship, and an $I(1)$ component, the product of the difference between the micro-coefficients and imposed common coefficient, and the $I(1)$ regressor. Thus the pooled regression will not constitute a cointegrating regression and the parameter estimates will not be consistent.

Consider now aggregating the micro-relations

$$y_{it} = \beta_i x_{it} + \varepsilon_{it}, \tag{3.1}$$

where y_{it} and x_{it} are each $I(1)$ and are pairwise cointegrated; namely ε_{it} is $I(0)$.[13] Under the random coefficient specification, $\beta_i = \beta + \eta_i$, $\eta_i \sim \text{iid}(0,\ \omega^2)$, the corresponding time-series aggregate model is given by

$$\bar{y}_t = \beta \bar{x}_t + \bar{v}_t, \tag{3.2}$$

$$\bar{v}_t = \bar{\varepsilon}_t + N^{-1} \sum_{i=1}^{N} \eta_i x_{it}. \tag{3.3}$$

Suppose now

$$x_{it} = x_{i,t-1} + u_{it}, \qquad u_{it} \sim (0, \tau_i^2),$$

and for expositional simplicity assume that x_{i0} are fixed. In this case we have

$$\operatorname{cov}(\bar{v}_t,\ \bar{v}_{t-s}) = \operatorname{cov}(\bar{\varepsilon}_t,\ \bar{\varepsilon}_{t-s}) + \frac{\omega^2 \tau^2}{N}(t-s), \qquad t > s,$$

and $V(\bar{v}_t) = V(\bar{\varepsilon}_t) + (\tau^2 \omega^2 / N) t$, where as before $\tau^2 = N^{-1} \sum_{i=1}^{N} \tau_i^2$. Hence, for a fixed N, $\{\bar{v}_t\}$ will eventually behave as a random walk and $\bar{y}_t$ and $\bar{x}_t$ need not be cointegrated, even though *all* the underlying micro-relations are by assumption cointegrated. For a further analysis of cointegration and aggregation see Gonzalo (1993), where the conditions under which cointegration at micro-levels implies cointegration at the macro level, and *vice versa*, are discussed.

3.1. Cross-section estimates with integrated variables

We now examine the impact of nonstationary variables on the cross-section estimates. Once again consider the micro-relations (3.1). Here for simplicity of exposition we are considering the case where x_{it} is a scalar variable. Averaging the data across time periods and assuming random coefficients as above the OLS regression from the cross-section yields

$$\hat{\beta}^c = \frac{\sum_{i=1}^{N} \tilde{y}_i \tilde{x}_i}{\sum_{i=1}^{N} \tilde{x}_i^2} = \beta + \frac{\sum_{i=1}^{N} \eta_i \tilde{x}_i^2}{\sum_{i=1}^{N} \tilde{x}_i^2} + \frac{\sum_{i=1}^{N} \tilde{\varepsilon}_i \tilde{x}_i}{\sum_{i=1}^{N} \tilde{x}_i^2}. \tag{3.4}$$

Suppose x_{it} is a random walk with drift

$$x_{it} = x_{i,t-1} + \mu_i + u_{it}, \qquad u_{it} \sim (0, \tau_i^2). \tag{3.5}$$

---

[13] An intercept term can be readily allowed for in this model by regarding y_{it} and x_{it} being measured as deviations from their initial values x_{i0} and y_{i0}, treating these as fixed and assuming that $\varepsilon_{i0} = 0$.

Then

$$x_{it} = x_{i0} + \mu_i t + \sum_{j=1}^{t} u_{ij},$$

and the time averages, measured as deviations from the initial values x_{i0}, are given by

$$\tilde{x}_i = \mu_i + \left(\frac{T+1}{2}\right) + \sum_{j=1}^{T} (T+1-j)u_{ij}/T,$$

and hence

$$\mathrm{E}\left(N^{-1}\sum_{i=1}^{N} \tilde{x}_i^2\right) = \left(\frac{T+1}{2}\right)^2 \left(\sum_{i=1}^{N} \mu_i^2/N\right) + \sum_{i=1}^{N}\sum_{j=1}^{T}\sum_{k=1}^{T} (T+1-j)(T+1-k)\,\mathrm{E}(u_{ij}u_{ik})/NT^2,$$

which under (3.5) yields

$$\mathrm{E}\left(N^{-1}\sum_{i=1}^{N} \tilde{x}_i^2\right) = \left(\sum_{i=1}^{N} \mu_i^2/N\right)\left(\frac{T+1}{2}\right)^2 + \frac{\tau^2}{T^2}\sum_{j=1}^{T} (T+1-j)^2.$$

Now taking limits as $N \to \infty$ for a fixed T, we have[14]

$$\lim_{N\to\infty} \mathrm{E}\left(\sum_{i=1}^{N} \tilde{x}_i^2/N\right) = \left(\frac{T+1}{2}\right)^2 \lim_{N\to\infty}\left(\sum_{i=1}^{N} \mu_i^2/N\right) + \frac{T(T+1)(2T+1)}{6T^2} \lim_{N\to\infty}\left(\sum_{i=1}^{N} \tau_i^2/N\right),$$

which is finite for a finite T. Also under the assumption that x_{it}'s are strictly exogenous we have

$$\mathrm{E}\left(\sum_{i=1}^{N} \tilde{x}_i^2 \eta_i/N\right) = 0, \qquad \mathrm{E}\left(\sum_{i=1}^{N} \tilde{x}_i^2 \tilde{\varepsilon}_i/N\right) = 0.$$

Hence as $N \to \infty$, $\hat{\beta}^c \xrightarrow{p} \beta$, for a finite T.

This result indicates that the spurious correlation problem does not arise in the case of cross-section regressions, even if the underlying variables x_{it} and y_{it} contain a unit root. Under the above assumptions $\hat{\beta}^c$ is also an unbiased estimator of β, though the usual standard error formulae for the estimates are no longer valid.

[14] Note that $\sum_{j=1}^{T} (T+1-j)^2 = T(T+1)(2T+1)/6$.

The disturbance term in the cross-section regression is given by

$$\tilde{v}_i = \tilde{y}_i - \beta\tilde{x}_i = \tilde{\varepsilon}_i + \eta_i\tilde{x}_i,$$

with variance

$$V(\tilde{v}_i\,|\,\boldsymbol{x}) = V(\tilde{\varepsilon}_i) + \omega^2\tilde{x}_i^2,$$

where ω^2 is the variance of η_i. The first term is likely to be small, particularly if T is large, thus consistent estimates of the conditional variances of the cross-section coefficients can be obtained using White's procedure. It is, however, important to note that the above results are valid under the rather strong assumption that x_{it}'s are strictly exogenous, and do not hold in the standard case discussed in the time-series literature of cointegrated variables where possible dependence of x_{it} on $\varepsilon_{it'}$ (for some t and t') is not ruled out.

4. An application: Labour demand functions

The various procedures discussed above will be illustrated by an application previously used to examine the aggregation problem in linear models in PPK (1989) and LPP (1990). Employment functions will be estimated for a panel of 38 industries over the 29 years, 1956–84.[15] The logarithm of industry employment (e_{it}), measured in man-hours, is explained by current and lagged values of the logarithms of industry output (y_{it}), real product wage (w_{it}), and aggregate output ($\bar{y}_t$); together with two lags of log employment, an intercept, and a time trend.

This data set is used for illustration, primarily because it has already been documented and analyzed in previous papers. But in the light of the theory developed above it has a particular feature that may also make it relevant to the analysis of other large N, large T panels. For almost every industry the hypothesis that the output, employment, and real wage variables have unit roots cannot be rejected according to Augmented Dickey–Fuller statistics with zero, one, and two lagged changes. Only for output in industry 37 (business services) is there a unanimous rejection of the unit root hypothesis. There is one other case (industry 18) where there is conflict between the test statistics for output. There is one case of conflict for employment and five industries where one or two of the ADF statistics suggest rejection of the unit root null for real wages.

The primary parameters of interest are the long-run elasticities with respect to real wages and output. In the case of output, since an industry's employment is affected by its own output as well as by aggregate output, there are two possible measures of the long-run output elasticity: the response to its own output only,

[15] The full data set has 41 industries. For comparability with PPK and LPP we exclude the oil industry, where the data are zero for half the period and shipbuilding and tobacco where the common specification chosen for all the industries did not yield sensible results.

or the total response to both its own and aggregate output. Since aggregate output at any moment in time does not vary over industries, the parameters that can be recovered differ between procedures. The aggregate time-series can only give an estimate of the elasticity with respect to all output. The cross-sections can only give an estimate of the elasticity with respect to own output, and the aggregate effect, being the same for all groups, gets picked up in the intercept.

The long-run coefficients can be estimated by four basic methods: group means, aggregate time-series, pooled, and cross-section.

4.1. Group mean estimates

In this case the average long-run coefficients can be obtained in three ways:
1) from the mean of the long-run industry-specific coefficients,
2) from the average of industry specific short-run coefficients,
3) from the mean coefficients in the industry-specific cointegrating regressions.

In each case the means can be unweighted/weighted using the Swamy (1971) GLS procedure. The various estimates together with their standard errors are given in Table 1. In the case of the cointegrating regressions the reported standard errors, which are based on inappropriate estimates of the variances are given only for completeness. Our assumption that between industry disturbance covariances are zero can be relaxed. More specifically, the equation estimated for each industry has the autoregressive distributed lag (ADRL) form

$$\begin{aligned} e_{it} = {} & \alpha_{1i} + \alpha_{2i}t + \beta_{1i}y_{it} + \beta_{2i}y_{i,t-1} + \beta_{3i}\bar{y}_t + \beta_{4i}\bar{y}_{t-1} \\ & + \gamma_{1i}w_{it} + \gamma_{2i}w_{i,t-1} + \lambda_{1i}e_{i,t-1} + \lambda_{2i}e_{it-2} + \varepsilon_{it}, \\ & i = 1,2,\ \dots,38, \quad t = 1956,\ \dots,1984, \end{aligned} \tag{4.1}$$

e_{it} is employment, y_{it} output, w_{it} real wage, and $\bar{y}_t$ aggregate output, all in logarithms. If we define the long-run elasticities for industry i as:

with respect to the real wage, $\hat{\phi}_i = (\hat{\gamma}_{1i} + \hat{\gamma}_{2i})/(1 - \lambda_{1i} - \lambda_{2i})$,

with respect to all output, $\hat{\theta}_{1i} = (\hat{\beta}_{1i} + \hat{\beta}_{2i} + \hat{\beta}_{3i} + \hat{\beta}_{4i})/(1 - \hat{\lambda}_{1i} - \hat{\lambda}_{2i})$,

and with respect to own output, $\hat{\theta}_{2i} = (\hat{\beta}_{1i} + \hat{\beta}_{2i})/(1 - \hat{\lambda}_{1i} - \hat{\lambda}_{2i})$,

the mean of the long runs (panel A(i) in Table 1) are computed as[16]

$$\hat{\phi} = \sum_{i=1}^{N} \hat{\phi}_i/N,$$

similarly for $\hat{\theta}_1$ and $\hat{\theta}_2$. The long-run estimates calculated from the means of the

[16]Notice that the estimators of the long-run parameters $\hat{\phi}_i$, $\hat{\theta}_{1i}$, $\hat{\theta}_{2i}$, and the mean group estimators based on them are also superconsistent when the regressors are $I(1)$. See Pesaran and Shin (1995).

Table 1
Heterogeneous panel estimates of mean long-run labour demand elasticities

	Wages ($\hat{\phi}$)	Output: Total ($\hat{\theta}_1$)	Output: Own ($\hat{\theta}_2$)
(A) Based on dynamic industry regressions Eq. (4.1)			
(i) Means of industry long-run estimates			
Unweighted	– 0.33	0.85	0.42
	(0.27)	(0.25)	(0.19)
Weighted (Swamy estimates)	– 0.36	0.83	0.58
	(0.22)	(0.21)	(0.14)
(ii) Long-runs from means of industry short-run estimates			
Unweighted	– 0.45	1.03	0.57
	(0.10)	(0.18)	(0.11)
Weighted (Swamy estimates)	– 0.45	0.99	0.61
	(0.11)	(0.19)	(0.11)
(B) Based on cointegrating industry regressions, Eq. (4.2)			
Unweighted	– 0.30	0.64	0.36
	(0.05)	(0.11)	(0.07)
Weighted (Swamy estimates)	– 0.31	0.68	0.39
	(0.04)	(0.10)	(0.06)

Based on separate regressions on 38 industries, 1956–84 ($T = 29$). The standard errors of the estimates are given in parentheses, and computed assuming that regression coefficients are independently distributed across industries.

short-run coefficients use

$$\bar{\gamma}_1 = \sum_{i=1}^{N} \hat{\gamma}_{1i}/N, \quad \bar{\gamma}_2 = \sum_{i=1}^{N} \hat{\gamma}_{2i}/N, \quad \bar{\lambda}_1 = \sum_{i=1}^{N} \hat{\lambda}_{1i}/N, \quad \text{etc.}$$

to give the estimates in panel A(ii), namely

$$\bar{\phi} = (\bar{\gamma}_1 + \bar{\gamma}_2)/(1 - \bar{\lambda}_1 - \hat{\lambda}_2).$$

The cointegrating regressions have the form

$$e_{it} = \alpha_{1i} + \alpha_{2i}t + \theta_{2i}y_{it} + (\theta_{1i} + \theta_{2i})\bar{y}_t + \phi_i w_{it} + u_{it}, \tag{4.2}$$
$$i = 1, 2, \ldots, 38, \quad t = 1956, \ldots, 1984,$$

and the average long-run effects are computed as

$$\bar{\phi} = \sum_{i=1}^{N} \hat{\phi}_i/N, \quad \bar{\theta}_1 = \sum_{i=1}^{N} \hat{\theta}_{1i}/N, \quad \text{etc.,}$$

given in panel B.

Table 2
Aggregate time-series estimates of long-run labour demand elasticities

	Wages ($\hat{\phi}$)	Output: Total ($\hat{\theta}_1$)	Output: Own ($\hat{\theta}_2$)
(A) Based on dynamic regression Eq. (4.3)	− 1.22 (0.34)	1.51 (0.30)	N/A
(B) Based on cointegrating regression Eq. (4.4)	− 0.33 (0.13)	0.71 (0.10)	N/A

Based on a single regression, over 1956–84 ($T = 29$) using the averages for each year over 38 industries. The standard errors of the estimates are given in parentheses.

It is clear from the results in Table 1 that the estimates of the average long-run effects do differ. The weighting has the largest influence on the mean of the long runs, A(i). The individual estimates of the long-run effect have a large dispersion and a large variation in the accuracy of their estimates. Whether all the estimates of (4.2) constitute cointegrating relations may be questioned. The mean and standard deviation of the ADF(1) statistics on the residuals were − 3.37 and 0.86, respectively. The individual industry equations showed little sign of misspecification. The full results can be found in PPK's Table 1. The results presented here differ a little from PPK, because they used restricted versions of the equations, whereas we used the unrestricted equation for all industries.

The estimates used to derive panel A(ii) are given in the first column of Table 4.

4.2. Aggregate time-series estimates

The results for the aggregate model are presented in Table 2. The aggregate equation was estimated in a full dynamic form:

$$\bar{e}_t = \alpha_1 + \alpha_2 t + \beta_3 \bar{y}_t + \beta_4 \bar{y}_{t-1} + \gamma_1 \bar{w}_t + \gamma_2 \bar{w}_{t-1} + \lambda_1 \bar{e}_{t-1} + \lambda_2 \bar{e}_{t-2} + \bar{v}_t, \tag{4.3}$$

Row A in Table 2 shows the long-run wage elasticity $\hat{\phi} = (\hat{\gamma}_1 + \hat{\gamma}_2)/(1 - \hat{\lambda}_1 - \hat{\lambda}_2)$ and the total output elasticity $\hat{\theta}_1 = (\hat{\beta}_3 + \hat{\beta}_4)/(1 - \hat{\lambda}_1 - \hat{\lambda}_2)$ (the own output elasticity not being identified). Row B gives the estimates from the cointegrating regression

$$\bar{e}_t = \alpha_1 + \alpha_2 t + \theta_1 \bar{y}_t + \phi \bar{w}_t + \bar{u}_t. \tag{4.4}$$

Again the hypothesis of cointegration might be questioned, since the ADF(1) statistic at − 3.23 would not quite reject the null that the residuals from the cointegrating regression are $I(1)$.

Table 3
Pooled estimates of long-run labour demand elasticities

		Output	
	Wages ($\hat{\phi}$)	Total ($\hat{\theta}_1$)	Own ($\hat{\theta}_2$)
(A) Based on dynamic regression, Eq. (4.6)[a]			
Assuming industry intercepts:			
(i) are the same	− 3.28	24.63	3.15
	(5.60)	(45.55)	(4.37)
(ii) differ randomly	− 1.83	10.18	1.68
	(1.52)	(9.06)	(0.82)
(iii) differ and are fixed	− 0.74	1.65	0.87
	(0.17)	(0.40)	(0.13)
(B) Based on static regression, Eq. (4.5)[a]			
Assuming industry intercepts:			
(i) are the same	− 0.53	0.84	0.87
	(0.04)	(0.28)	(0.02)
(ii) differ randomly	− 0.42	0.76	0.61
	(0.02)	(0.05)	(0.02)
(iii) differ and are fixed	− 0.41	0.76	0.60
	(0.02)	(0.05)	(0.02)
(C) Based on first-differenced regression[b]			
(i) Static[c]	− 0.24	0.44	0.28
	(0.02)	(0.04)	(0.02)
(ii) Dynamics[d]	− 0.35	1.02	0.56
	(0.03)	(0.08)	(0.04)
(iii) Dynamic IV[e]	− 0.36	1.08	0.58
	(0.05)	(0.10)	(0.05)

The standard errors of the estimates are given in parentheses.

[a]Using 1102 observations, $t = 1956$–84 ($T = 29$) with $N = 38$.

[b]Using 1026 observations, $t = 1958$–84 ($T = 27$) with $N = 38$.

[c]Δe_{it} on Δy_{it}, $\Delta \bar{y}_t$, Δw_{it}, and an intercept.

[d]As footnote c plus $\Delta y_{i,t-1}$, $\Delta \bar{y}_{t-1}$, $\Delta w_{i,t-1}$, $\Delta e_{i,t-1}$, $\Delta e_{i,t-2}$.

[e]The same specification as in footnote d, using Δy_{it}, $\Delta \bar{y}_t$, Δw_{it}, $\Delta y_{i,t-1}$, $\Delta \bar{y}_{t-1}$, $\Delta w_{i,t-1}$, $\Delta y_{i,t-2}$, $\Delta \bar{y}_{t-2}$, $\Delta w_{i,t-2}$, $e_{i,t-3}$, $e_{i,t-4}$, and ones as instruments.

It is noticeable that the aggregate 'cointegrating' regression gives results close to the means of the cointegrating micro-regression. This is what we would expect from the Zellner (1969) argument, given that no lagged dependent variables are included and the coefficients appear to be independent of the

Table 4
Alternative estimates of the dynamic labour demand equation

	Swamy estimates[a]	Pooled estimates[b]	Anderson–Hsiao estimates[c]
Intercept	− 0.59 (0.73)	—	—
y_{it}	0.29 (0.04)	0.24 (0.02)	0.25 (0.02)
$y_{i,t-1}$	− 0.07 (0.05)	− 0.20 (0.02)	0.11 (0.04)
w_{it}	− 0.23 (0.03)	− 0.23 (0.01)	− 0.23 (0.01)
$w_{i,t-1}$	0.07 (0.03)	0.20 (0.02)	0.01 (0.04)
$\bar{y}_t$	0.14 (0.07)	0.16 (0.04)	0.18 (0.04)
$\bar{y}_{t-1}$	− 0.001 (0.06)	− 0.12 (0.04)	0.12 (0.05)
$e_{i,t-1}$	0.79 (0.06)	1.16 (0.03)	0.22 (0.14)
$e_{i,t-2}$	− 0.14 (0.05)	− 0.21 (0.03)	0.16 (0.09)
t (coefficient × 100)	− 0.60 (0.17)	0.15 (0.03)	− 1.33 (0.17)

The standard errors of the estimates are given in parentheses.
[a]Described in Table 1, panel A(ii).
[b]Described in Table 3, panel A(iii), assuming intercepts are different and fixed.
[c]Described in Table 3, panel C(iii).

regressors. As the results in PPK showed, the aggregate estimates are a very long way from the average of the industry coefficients. Given that the regressors are positively serially correlated, aggregation induces positive serial correlation in the residuals of the aggregate equation. We might expect this to bias the coefficient of the lagged dependent variable upwards, exaggerating the estimate of the long-run effect. The coefficients of the lagged dependent variables are 0.49 and 0.16 in the aggregate and 0.79 and − 0.14 in the case of the Swamy estimates, so the sum is close though the pattern is not. It is also noticeable when comparing the aggregate with the Swamy estimators, that $\bar{w}_{t-1}$ and $\bar{y}_{t-1}$, in addition to $\bar{e}_{t-2}$ have opposite signs. Thus the impression of the dynamics given by the aggregate and averages are quite different. In addition, $\bar{y}_{t-1}$ and $y_{i,t-1}$ are insignificant in the average, but significant in the aggregate.

4.3. The pooled estimates

Table 3 provides estimates on the basis of the pooled regression. A static version is estimated, for comparison with the cointegrating regressions and the cross-sections (set B),

$$e_{it} = \alpha_{1i} + \alpha_2 t + \theta_2 y_{it} + (\theta_1 - \theta_2)\bar{y}_t + \phi w_{it}, \tag{4.5}$$

where the long-run coefficients are just θ_1, θ_2, and ϕ. Then a dynamic version is estimated (set A),

$$\begin{aligned} e_{it} = {} & \alpha_{1i} + \alpha_2 t + \beta_1 y_{it} + \beta_2 y_{it-1} + \beta_3 \bar{y}_t + \beta_4 \bar{y}_{t-1} \\ & + \gamma_1 w_{it} + \gamma_2 w_{i,t-1} + \lambda_1 e_{i,t-1} + \lambda_2 e_{i,t-2}, \end{aligned} \tag{4.6}$$

where the long-run coefficients are

$$\hat{\phi} = (\hat{\gamma}_1 + \hat{\gamma}_2)/(1 - \hat{\lambda}_1 - \hat{\lambda}_2), \quad \hat{\theta}_1 = (\hat{\beta}_1 + \hat{\beta}_2)/(1 - \hat{\lambda}_1 - \hat{\lambda}_2), \quad \text{etc.}$$

For each set three estimators are used which differ in the assumption about α_{1i}:

$\alpha_{1i} = \alpha_1$,	constant,
$\alpha_{1i} = \alpha_1 + \mu_{1i}$,	group-specific random effects, where μ_{1i} are the random components,
$\alpha_{1i} = \alpha_{1i}$,	group-specific fixed effects.

The most obvious feature of the results is the erratic performance of the dynamic pooled model, whose estimates are very sensitive to specification. The estimates for the dynamic model with fixed industry effects are given in Table 4. Just as the theory predicts, the sum of the coefficients of the lagged dependent variable is very close to unity and the coefficients of the lagged exogenous variables are roughly equal and opposite in sign to the coefficients of their current values. Thus the estimate of the long-run coefficients are the ratios of two numbers which are tending to zero, and are likely to be computationally unstable. For example, the estimator of the income elasticity for the pooled estimates with industry-specific dummies is $0.04004/(1 - 0.95406) = 0.87$ with a standard error of 0.13. Although this is not too far from the Swamy estimator of the long run from means 0.61 (0.11), the Swamy estimates suggest that the denominator should be 0.36 rather than 0.05 and the numerator 0.22 rather than 0.04. Note also that the fact that the pooled result looks like a first difference equation cannot be attributed to the need to remove group-specific intercepts, since these are included.

Although the fixed effect model is inconsistent with lagged dependent variables for fixed T even if N goes to infinity, this should not be a problem here, since T is relatively large. To investigate this effect, the model was also estimated using the Anderson–Hsiao procedure (e.g., Hsiao, 1986, p. 89). The dynamic equation is differenced to remove the fixed effects and then estimated by Instrumental Variables (IV) to deal with the correlation between the disturbances and the regressors caused by the induced MA(1) error. The instruments used were the current and two lags of the changes in the exogenous variables and the third and fourth lags of the dependent variable. Use of the extra two lags reduces the sample to 27×38 observations. The estimates of the long-run coefficients are given in Table 3 and the full results in Table 4. The differenced equation was also estimated by OLS which, as can be seen from Table 3, gives almost exactly the same results, and was estimated without the lags. When the equation is differenced before estimation, some of the serial correlation induced by the inappropriate imposition of homogeneity on the slope coefficients is removed. The coefficients on the current exogenous variables are very similar to the pooled and Swamy estimates as the theory predicts, but they do not suffer such extreme biases in the coefficients of the lagged exogenous variables as do the undifferenced pooled estimates. The coefficient of the lagged dependent variable seems to be seriously underestimated.

4.4. *Cross-section estimates*

The cross-section estimates using means for each industry over all 29 years are

$$\begin{array}{llll} \tilde{e} = & -\,1.273 & +\,0.876\,\tilde{y}_i & -\,0.523\,\tilde{w}_i, \\ & (1.675) & (0.106) & (0.266) \\ & [2.653] & [0.103] & [0.439] \end{array}$$

$$R^2 = 0.71, \qquad \hat{\sigma} = 0.641, \qquad N = 38.$$

The OLS standard errors of the estimates are given in parentheses, White's heteroscedasticity-consistent standard errors in brackets. The argument at the end of Section 3.1 suggests that the latter are appropriate. Note that the output coefficient is an estimate of the own elasticity θ_2. The cross-section was also estimated recursively, using first the data for 1954, then the averages for 1954 and 1955, extending the window till all 29 years were included. In this case the estimates seem quite robust to variations in the number of years used to form averages. The output elasticities all lie between 0.84 and 0.88, the wage elasticities between -0.58 and -0.42.

In this particular application, the cross-section provides quite reasonable point estimates of the mean of the long-run coefficients, although these estimates are rather poorly determined. However, it is important to be clear about the conditions under which this will be true. The coefficients and regressors appear to be independent, and the cross-section only provides an estimate of the own-output effect, not the total effect. Where macro-influences on individual decisions are important, because of interaction effects between groups for instance, the cross-section estimates will not reveal them.

5. Conclusions

The main conclusion of this paper is that the random coefficient argument of Zellner (1969), which seems to have been widely accepted in the literature, does not extend to dynamic models. Thus aggregating or pooling dynamic heterogeneous panels can produce very misleading estimates, as our applied example illustrates. Given the prevalence of aggregation and pooling in applied work, these results are of some importance, and indicate that the common assumption of homogeneity in dynamic models is far from innocuous. Our theoretical and empirical results provided some qualified support for the belief that cross-section estimates will provide estimates of long-run effects, which are robust to dynamic misspecification of the underlying time-series model. One can only make limited inferences from our particular applied example because there are specific features of the data which are important. In particular, in our application, it seems that the regressors are $I(1)$, the variables cointegrate, and the coefficients are independent of the regressors. All these features are important for the results. As the theory indicated the pooled and aggregated time-series regressions can give very misleading results for the dynamic model. First differencing appeared to reduce the problem somewhat for the pooled model, since it seemed to remove the serial correlation which was the primary source of the inconsistency, but it is not clear how general this result is. Simple static regressions also did quite well, perhaps because the argument for random coefficients with exogenous regressors can be approximately applied to the cointegrating regressions characteristic of this data, but our theoretical results showed this will not, in general, be true.

The lesson for applied work is that when large T panels are available, the individual micro-relations should be estimated separately and the averages of the estimated micro-parameters and their standard errors calculated explicitly. With modern computational facilities this is very easy to do. The hypothesis of homogeneity, common slope coefficients, can then be tested. Our experience is that it is almost always rejected, even when the size of the test is adjusted to take account of the large number of observations. The statistical relationship between the coefficients and the group means can then be examined. If coefficients are related to explanatory variables, which they were not in our data, these will cause the cross-section estimates to differ from the average of the time-series. Such correlated effects can then be investigated. Our applied example also brought out the importance of being explicit about variables which differ over time but not over groups, like aggregate output, and which differ across groups but not over time.

Although we have tried to be fairly comprehensive, we have only been able to present a representative selection of results, there are a range of other specifications of the model which could be considered. For instance, our results on the consistency of the cross-section estimates and cointegration relations required the regressors to be strictly exogenous, a much stronger assumption than is usually employed. The assumption of independence of the coefficients and regressors is crucial and appropriate tests for this assumption are required.

Appendix A

Inconsistency of the pooled estimators in the context of a heterogeneous dynamic panel

Consider the heterogeneous dynamic model (2.1), explicitly including a random intercept,

$$y_{it} = \alpha_i + \lambda_i y_{i,t-1} + \beta_i x_{it} + \varepsilon_{it}, \qquad i = 1,2,\ldots,N, \quad t = 1,2,\ldots,T, \tag{A.1}$$

and suppose that the parameters λ_i and β_i are random with means λ and β, respectively, variances ω_{11} and ω_{22}, and the covariance $\omega_{12} = \omega_{21}$. To simplify the exposition we consider the case where β_i is a scalar coefficient, and the process generating x_{it} can be characterized by the simple stationary AR(1) process

$$x_{it} = \mu_i(1-\rho) + \rho x_{i,t-1} + u_{it},$$

where μ_i is the (unconditional) mean of x_{it}, $|\rho| < 1$, and for each i, $u_{it} \sim \text{iid}(0, \tau_i^2)$.

Suppose that λ and β are estimated by the fixed-effect (or 'within') estimators using the pooled regression

$$y_{it} = \alpha_i + \lambda y_{i,t-1} + \beta x_{it} + v_{it}. \tag{A.2}$$

Here we derive the probability limits of the fixed-effect estimators of λ and β, which we denote by $\hat{\lambda}$ and $\hat{\beta}$, in the case where v_{it} is given by

$$v_{it} = \varepsilon_{it} + \eta_{1i} y_{i,t-1} + \eta_{2i} x_{it}. \tag{A.3}$$

This formulation allows for the randomness in the slope coefficients through the terms involving η_{1i} and η_{2i}, the random components of λ_i and β_i, respectively. In deriving the probability limits we first let $T \to \infty$ and then let $N \to \infty$. The same limiting expressions obtain if this order is reversed.

Using standard notations, the fixed effect estimators of λ and β in (A.2) are given by

$$\begin{pmatrix} \hat{\lambda} \\ \hat{\beta} \end{pmatrix} = \begin{pmatrix} \boldsymbol{y}'_{-1} \boldsymbol{W}_n \boldsymbol{y}_{-1} & \boldsymbol{y}'_{-1} \boldsymbol{W}_n \boldsymbol{x} \\ \boldsymbol{x}' \boldsymbol{W}_n \boldsymbol{y}_{-1} & \boldsymbol{x}' \boldsymbol{W}_n \boldsymbol{x} \end{pmatrix}^{-1} \begin{pmatrix} \boldsymbol{y}'_{-1} \boldsymbol{W}_n \boldsymbol{y} \\ \boldsymbol{x}' \boldsymbol{W}_n \boldsymbol{y} \end{pmatrix}, \tag{A.4}$$

where $\boldsymbol{y}$ and $\boldsymbol{x}$ are $NT \times 1$ vectors of observations on y_{it}'s and x_{it}'s, $\boldsymbol{y}_{-1}$ is $NT \times 1$ vector of observations on $y_{i,t-1}$, $i = 1, 2, \ldots, N$, $t = 1, 2, \ldots, T$. $\boldsymbol{W}_n$ is the 'within' transformation matrix operator,

$$\boldsymbol{W}_n = \boldsymbol{I}_{NT} - \boldsymbol{D}(\boldsymbol{D}(\boldsymbol{D}'\boldsymbol{D})^{-1}\boldsymbol{D}', \tag{A.5}$$

where

$$\boldsymbol{D} = \boldsymbol{I}_N \otimes \tau_T, \tag{A.6}$$

$\boldsymbol{I}_N$ is the identity matrix, τ_T is a $T \times 1$ unit vector, $\tau_T = (1, 1, \ldots, 1)'$, and $\otimes$ stands for matrix Kroneker product (see, for example, Chapter 4 in Matyas and Sevestre, 1992). Under (A.2) and (A.3) we have

$$\begin{pmatrix} \hat{\lambda} - \lambda \\ \hat{\beta} - \beta \end{pmatrix} = \begin{Bmatrix} \dfrac{\boldsymbol{y}'_{-1}\boldsymbol{W}_n\boldsymbol{y}_{-1}}{NT} & \dfrac{\boldsymbol{y}'_{-1}\boldsymbol{W}_n\boldsymbol{x}}{NT} \\ \dfrac{\boldsymbol{x}'\boldsymbol{W}_n\boldsymbol{y}_{-1}}{NT} & \dfrac{\boldsymbol{x}'\boldsymbol{W}_n\boldsymbol{x}}{NT} \end{Bmatrix}^{-1} \begin{Bmatrix} \dfrac{\boldsymbol{y}'_{-1}\boldsymbol{W}_n\boldsymbol{v}}{NT} \\ \dfrac{\boldsymbol{x}'\boldsymbol{W}_n\boldsymbol{v}}{NT} \end{Bmatrix}, \tag{A.7}$$

where $\boldsymbol{v}$ is $NT \times 1$ vector of the 'composite errors' with typical elements given by (A.3). We have $\boldsymbol{v} = (\boldsymbol{v}'_1, \boldsymbol{v}'_2, \ldots, \boldsymbol{v}'_N)'$, where $\boldsymbol{v}_i$ are $T \times 1$ vectors,

$$\boldsymbol{v}_i = \varepsilon_i + \eta_{1i}\boldsymbol{y}_{i,-1} + \eta_{i2}\boldsymbol{x}_i. \tag{A.8}$$

Now using $\boldsymbol{W}_n$ (see (A.5)), it is easily seen that

$$\boldsymbol{x}'\boldsymbol{W}_n\boldsymbol{x} = \sum_{i=1}^{N} \boldsymbol{x}'_i\boldsymbol{H}_T\boldsymbol{x}_i,$$

where

$$\boldsymbol{H}_T = \boldsymbol{I}_T - \tau_T(\tau'_T\tau_T)^{-1}\tau'_T.$$

Similarly,

$$\boldsymbol{y}'_{-1}\boldsymbol{W}_n\boldsymbol{y}_{-1} = \sum_{i=1}^{N} \boldsymbol{y}'_{i,-1}\boldsymbol{H}_T\boldsymbol{y}_{i,-1}, \quad \boldsymbol{x}'\boldsymbol{W}_n\boldsymbol{y}_{-1} = \sum_{i=1}^{N} \boldsymbol{x}'_i\boldsymbol{H}_T\boldsymbol{y}_{i,-1}, \quad \text{etc.}$$

This type of decomposition now allows us, for example, to write

$$\operatorname*{plim}_{N\to\infty,\, T\to\infty} \left(\frac{\boldsymbol{x}'\boldsymbol{W}_n\boldsymbol{x}}{NT}\right) = \operatorname*{plim}_{N\to\infty} \left\{\frac{1}{N}\sum_{i=1}^{N} \operatorname*{plim}_{T\to\infty}\left(\frac{\boldsymbol{x}'_i\boldsymbol{H}_T\boldsymbol{x}_i}{T}\right)\right\}, \tag{A.9}$$

where the limits are taken first letting $T \to \infty$ and then $N \to \infty$. As another example, note that

$$\operatorname*{plim}_{N \to \infty,\, T \to \infty} \left(\frac{y'_{-1} W_n v}{NT} \right) = \operatorname*{plim}_{N \to \infty} \left\{ \frac{1}{N} \sum_{i=1}^{N} \operatorname*{plim}_{T \to \infty} \left(\frac{y'_{i,-1} H_T v_i}{T} \right) \right\}, \tag{A.10}$$

To obtain the probability limits of the various terms in (A.7) we first write (A.1) in vector notations as

$$y_i = \alpha_i \tau_T + \lambda_i y_{i,-1} + x_i \beta_i + \varepsilon_i,$$

and under Assumption 3 in the text, and assuming that the process (A.1) generating y_{it} has started a long time ago we have

$$y_i = \left(\frac{\alpha_i}{1 - \lambda_i} \right) \tau_T + \sum_{j=0}^{\infty} \lambda_i^j \beta_i x_{i,-j} + \sum_{j=0}^{\infty} \lambda_i^j \varepsilon_{i,-j}, \tag{A.11}$$

where $x_{i,-j}$ and $\varepsilon_{i,-j}$ are $T \times 1$ vector of observations on the jth-order lags of x_i and ε_i.

Consider now the probability limit of $(y'_{-1} W_n^m v)/NT$. First using (A.8) in (A.10) we have

$$\operatorname*{plim}_{T \to \infty} \left(\frac{y'_{i,-1} H_T v_i}{T} \right) = \operatorname*{plim}_{T \to \infty} \left(\frac{y'_{i,-1} H_T \varepsilon_i}{T} \right) + \eta_{1i} \operatorname*{plim}_{T \to \infty} \left(\frac{y'_{i,-1} H_T y_{i,-1}}{T} \right)$$
$$+ \eta_{2i} \operatorname*{plim}_{T \to \infty} \left(\frac{y_{i,-1} H_T x_i}{T} \right). \tag{A.12}$$

Substituting from (A.11) in the above expressions and taking probability limits under Assumptions 1–4 (with $T \to \infty$) we have

$$\operatorname*{plim}_{T \to \infty} \left(\frac{y'_{i,-1} H_T \varepsilon_i}{T} \right) = \left(\frac{\alpha_i}{1 - \lambda_i} \right) \operatorname*{plim}_{T \to \infty} \left(\frac{\tau'_T H_T \varepsilon_i}{T} \right)$$
$$+ \sum_{j=0}^{\infty} \lambda_i^j \beta_i \operatorname*{plim}_{T \to \infty} \left(\frac{x'_{i,-j-1} H_T \varepsilon_i}{T} \right)$$
$$+ \sum_{j=0}^{\infty} \lambda_i^j \operatorname*{plim}_{T \to \infty} \left(\frac{\varepsilon'_{i,-j-1} H_T \varepsilon_i}{T} \right). \tag{A.13}$$

But $H_T \tau_T = 0$, and under Assumption 1 (in the text)

$$\operatorname*{plim}_{T \to \infty} \left(\frac{x'_{i,-j-1} H_T \varepsilon_i}{T} \right) = 0, \quad \forall i. \tag{A.14}$$

Also

$$T^{-1}(\varepsilon'_{i,-j}\boldsymbol{H}_T\varepsilon_i) = T^{-1}\sum_{t=1}^{T}(\varepsilon_{it}-\bar{\varepsilon}_{i0})'(\varepsilon_{i,t-j}-\bar{\varepsilon}_{i,-j}),$$

where $\bar{\varepsilon}_{i,-j} = T^{-1}\sum_{t=1}^{T}\varepsilon_{i,t-j}$ are time averages. Hence using results in Appendix C we also have

$$\operatorname*{plim}_{T\to\infty}\left(\frac{\varepsilon'_{i,-j}\boldsymbol{H}_T\varepsilon_i}{T}\right) = 0, \qquad j = 1,2,\ldots, \quad i = 1,2,\ldots,N. \tag{A.15}$$

$$\operatorname*{plim}_{T\to\infty}\left(\frac{\varepsilon'_{i}\boldsymbol{H}_T\varepsilon_i}{T}\right) = \sigma_i^2, \qquad i = 1,2,\ldots,N, \tag{A.16}$$

where $\sigma_i^2 = V(\varepsilon_{it})$. Using (A.14) and (A.15) in (A.13), we therefore have

$$\operatorname*{plim}_{T\to\infty}\left(\frac{\boldsymbol{y}'_{i,-1}\boldsymbol{H}_T\varepsilon_i}{T}\right) = 0. \tag{A.17}$$

Similarly, and after some algebra it can be shown that

$$\operatorname*{plim}_{T\to\infty}\left(\frac{\boldsymbol{y}'_{i,-1}\boldsymbol{H}_T\boldsymbol{y}_{i,-1}}{T}\right) = \left(\frac{\sigma_i^2}{1-\lambda_i^2}\right) + \sum_{j'=0}^{\infty}\sum_{j=0}^{\infty}\frac{\lambda_i^{j+j'}\beta_i^2\rho^{|j-j'|}\tau_i^2}{1-\rho^2}, \tag{A.18}$$

$$\operatorname*{plim}_{T\to\infty}\left(\frac{\boldsymbol{y}'_{i,-1}\boldsymbol{H}_T\boldsymbol{x}_i}{T}\right) = \sum_{j=0}^{\infty}\frac{\lambda_i^j\rho^{j+1}\beta_i\tau_i^2}{1-\rho^2}. \tag{A.19}$$

Substituting (A.17)–(A.19) back in (A.12) we have

$$\operatorname*{plim}_{T\to\infty}\left(\frac{\boldsymbol{y}'_{i,-1}\boldsymbol{H}_T\boldsymbol{v}_i}{T}\right) = \left(\frac{\sigma_i^2\eta_{1i}}{1-\lambda_i^2}\right) + \left(\frac{\tau_i^2\eta_{1i}\beta_i^2}{1-\rho^2}\right)\sum_{j'=0}^{\infty}\sum_{j=0}^{\infty}\lambda_i^{j+j'}\rho^{|j-j'|}$$
$$+ \left(\frac{\tau_i^2\eta_{2i}\beta_i}{1-\rho^2}\right)\sum_{j=0}^{\infty}\lambda_i^j\rho^{j+1}.$$

Finally, substituting this result in (A.10) yields

$$\operatorname*{plim}_{N\to\infty,\,T\to\infty}\left(\frac{\boldsymbol{y}'_{-1}\boldsymbol{W}_n\boldsymbol{v}}{NT}\right) = \operatorname*{plim}_{N\to\infty}\left\{\frac{1}{N}\sum_{i=1}^{N}\left(\frac{\sigma_i^2\eta_{1i}}{1-\lambda_i^2}\right)\right\}$$
$$+ \sum_{j'=0}^{\infty}\sum_{j=0}^{\infty}\frac{\rho^{|j-j'|}}{1-\rho^2}\operatorname*{plim}_{N\to\infty}\left(\frac{1}{N}\sum_{i=1}^{N}\eta_{1i}\tau_i^2\beta_i^2\lambda_i^{j+j'}\right)$$
$$+ \sum_{j=0}^{\infty}\frac{\rho^{j+1}}{1-\rho^2}\operatorname*{plim}_{T\to\infty}\left(\frac{1}{N}\sum_{i=1}^{N}\beta_i\eta_{2i}\tau_i^2\lambda_i^j\right),$$

and since by assumption λ_i, β_i, σ_i^2 and τ_i^2 are identically and independently distributed across i, and have moments of all orders, (see Assumption 3) by the law of large numbers, the probability limits as $N \to \infty$ converge to their respective means, and we have

$$\underset{N\to\infty,\,T\to\infty}{\text{plim}} \left(\frac{y'_{-1} W_n v}{NT}\right) = \mathrm{E}\left(\frac{\sigma_i^2 \eta_{1i}}{1-\lambda_i^2}\right) + \sum_{j'=0}^{\infty} \sum_{j=0}^{\infty} \frac{\rho^{|j-j'|}}{1-\rho^2} \mathrm{E}(\eta_{1i} \tau_i^2 \beta_i^2 \lambda_i^{j+j'}) + \sum_{j=0}^{\infty} \frac{\rho^{j+1}}{1-\rho^2} \mathrm{E}\left(\beta_i \eta_{2i} \tau_i^2 \lambda_i^j\right), \tag{A.20}$$

where the expectations are taken with respect to the joint distribution of η_{1i}, η_{2i}, τ_i^2, and σ_i^2. Recall that $\lambda_i = \lambda + \eta_{1i}$ and $\beta_i = \beta + \eta_{2i}$. The inconsistency of the fixed-effect estimators of λ and β in a dynamic heterogeneous model now follows from the fact that the above probability limit is, in general, nonzero even for N and T large. It vanishes only in the case where $\rho = 0$ and $\lambda_i = \lambda$ for all i (and hence $\eta_{1i} = 0$).[17]

Turning to the other terms in (A.7) and following a similar approach as above, we have

$$\underset{N\to\infty,\,T\to\infty}{\text{plim}} \left(\frac{x' W_n v}{NT}\right) = \sum_{j=0}^{\infty} \left(\frac{\rho^{j+1}}{1-\rho^2}\right) \mathrm{E}(\eta_{1i} \beta_i \tau_i^2 \lambda_i^j), \tag{A.21}$$

$$\underset{N\to\infty,\,T\to\infty}{\text{plim}} \left(\frac{x' W_n x}{NT}\right) = \frac{\mathrm{E}(\tau_i^2)}{1-\rho^2}, \tag{A.22}$$

and using (A.18) and (A.19)

$$\underset{N\to\infty,\,T\to\infty}{\text{plim}} \left(\frac{y'_{-1} W_n y_{-1}}{NT}\right) = \mathrm{E}\left(\frac{\sigma_i^2}{1-\lambda_i^2}\right) + \sum_{j'=0}^{\infty} \sum_{j=0}^{\infty} \frac{\rho^{|j-j'|}}{1-\rho^2} \mathrm{E}(\tau_i^2 \beta_i^2 \lambda_i^{j+j'}), \tag{A.23}$$

$$\underset{N\to\infty,\,T\to\infty}{\text{plim}} \left(\frac{y'_{-1} W_n x}{NT}\right) = \sum_{j=0}^{\infty} \left(\frac{\rho^{j+1}}{1-\rho^2}\right) \mathrm{E}(\beta_i \tau_i^2 \lambda_i^j). \tag{A.24}$$

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[17] Notice that even this result depends on the assumed homogeneity of the autoregressive coefficient of the $\{x_{it}\}$ process for all i.

Using the probability limits (A.20) to (A.24) in (A.7), *explicit* expressions for the inconsistency of the fixed-effect estimators can now be readily obtained. The resultant expressions, however, would still be rather complicated and for pedagogic purposes it is worth considering the relatively simple case where $\lambda_i = \lambda$ for all. In this case, apart from the intercept terms, the only remaining source of parameter variations across groups is through β_i. Using (A.20)–(A.24) and setting $\eta_{1i} = 0$, and $\lambda_i = \lambda$ for all i, and denoting the variance of β_i by ω_{22} we have[18]

$$\operatorname*{plim}_{N\to\infty,\, T\to\infty} \begin{pmatrix} \hat{\lambda} - \lambda \\ \hat{\beta} - \beta \end{pmatrix} = \begin{bmatrix} \dfrac{\sigma^2}{1-\lambda^2} + \dfrac{\delta\tau^2(\beta^2 + \omega_{22})}{1-\rho^2} & \dfrac{\beta\rho\tau^2}{(1-\rho^2)(1-\lambda\rho)} \\ \dfrac{\beta\rho\tau^2}{(1-\rho^2)(1-\lambda\rho)} & \dfrac{\tau^2}{1-\rho^2} \end{bmatrix}^{-1}$$

$$\times \begin{bmatrix} \dfrac{\rho\omega_{22}\tau^2}{(1-\rho^2)(1-\lambda\rho)} \\ 0 \end{bmatrix}, \tag{A.25}$$

which in turn yield (2.7) and (2.8) in the text.

Appendix B

Error properties of aggregate time-series models obtained from dynamic heterogeneous panels

Consider the micro-relations

$$y_{it} = \lambda_i y_{i,t-1} + \beta_i x_{it} + \varepsilon_{it}, \qquad i = 1,2,\ldots,N, \quad t = 1,2,\ldots,T, \tag{B.1}$$

where

$$\lambda_i = \lambda + \eta_{1i}, \qquad \beta_i = \beta + \eta_{2i},$$

[18] Variations in σ_i^2 and τ_i^2 across groups does not pose any difficulties so long as they are assumed to be independently distributed of β_i (and of λ_i, when λ_i is random). The same results also follow when $\sigma_i^2 = \sigma^2$, $\tau_i^2 = \tau^2$, for all i.

and Assumptions 1–3 in the text hold. Also assume for simplicity that x_{it}'s have zero means and that ε_{it}'s are serially uncorrelated. Here to simplify the exposition we shall confine the analysis to the case of a scalar x_{it}. The aggregate model associated with the above micro-specifications is given by

$$\bar{y}_t = \lambda \bar{y}_{t-1} + \beta \bar{x}_t + \bar{v}_t, \tag{B.2}$$

where

$$\bar{v}_t = \bar{\varepsilon}_t + N^{-1} \sum_{i=1}^{N} (\eta_{1i} x_{it} + \eta_{2i} y_{i,t-1}). \tag{B.3}$$

We now show that the aggregate time-series estimates of λ and β obtained by the OLS regression of $\bar{y}_t$ on $\bar{y}_{t-1}$ and $\bar{x}_t$ will be biased, and that this bias does not disappear even if both N and T tend to infinity.

The OLS estimator of β in (B.2), which we denote by $\hat{\beta}^a$, is given by

$$\hat{\beta}^a - \beta = \frac{\left(\sum_{t=1}^{T} \bar{y}_{t-1}^2\right)\left(\sum_{t=1}^{T} \bar{x}_t \bar{v}_t\right) - \left(\sum_{t=1}^{T} \bar{x}_t \bar{y}_{t-1}\right)\left(\sum_{t=1}^{T} \bar{y}_{t-1} \bar{v}_t\right)}{\left(\sum_{t=1}^{T} \bar{x}_t^2\right)\left(\sum_{t=1}^{T} \bar{y}_{t-1}^2\right) - \left(\sum_{t=1}^{T} \bar{x}_t \bar{y}_{t-1}\right)^2}. \tag{B.4}$$

Under Assumptions 1–3 and using (3.6) we have

$$\mathrm{E}(\bar{x}_t \bar{v}_t) = N^{-1} \sum_{i=1}^{N} \sum_{s=1}^{\infty} \mathrm{E}(\eta_{2i} \beta_i \lambda_i^s) \mathrm{E}(\bar{x}_t x_{i,t-s-1})$$

and

$$\mathrm{E}(\bar{y}_{t-1} \bar{v}_t) = N^{-1} \sum_{i=1}^{N} \{\mathrm{E}(\eta_{1i} x_{it} \bar{y}_{t-1}) + \mathrm{E}(\eta_{2i} y_{i,t-1} \bar{y}_{t-1})\},$$

where

$$\mathrm{E}(\eta_{1i} x_{it} \bar{y}_{t-1}) = N^{-1} \sum_{s=0}^{\infty} \mathrm{E}(\eta_{1i} \beta_i \lambda_i^s) \mathrm{E}(\bar{x}_{it} x_{i,t-s-1}),$$

$$\mathrm{E}(\eta_{2i} y_{i,t-1} \bar{y}_{t-1}) = N^{-1} \sum_{j=1}^{N} \mathrm{E}(\eta_{2i} y_{i,t-1} y_{j,t-1}),$$

where

$$\mathrm{E}(\eta_{2i} y_{i,t-1} \lambda_{j,t-1}) = \sum_{s=0}^{\infty} \sum_{s'=0}^{\infty} \mathrm{E}(\eta_{2i} \beta_i \lambda_i^s) \mathrm{E}(\beta_j \lambda_j^{s'}) \mathrm{E}(x_{i,t-s-1} x_{j,t-s'-1})$$

$$+ \sum_{s=0}^{\infty} \mathrm{E}(\eta_{2i} \lambda_i^s) \mathrm{E}(\lambda_j^s) \mathrm{E}(\varepsilon_{i,t-s-1} \varepsilon_{j,t-s-1}).$$

Moreover,

$$\mathrm{E}(\eta_{ji}\beta_i\lambda_i^s) = \mathrm{E}[\eta_{ji}(\beta + \eta_{2i})(\lambda + \eta_{1i})^s], \qquad j = 1, 2,$$

and $\mathrm{E}(\eta_{2i}\lambda_i^s) = \mathrm{E}[\eta_{2i}(\lambda + \eta_{1i})^s]$ are all nonzero and, therefore, under the standard assumptions of the random coefficient model the regressors in (B.2) and the composite aggregate disturbance term $\bar{v}_t$ will be correlated. In the absence of parameter homogeneity, the correlation between $\bar{x}_t$ and $\bar{v}_t$ vanishes only if x_{it}'s are serially uncorrelated. The conditions under which $\mathrm{E}(\bar{v}_t\bar{y}_{t-1}) = 0$ are more stringent and also require that x_{it} and ε_{it} to be distributed independently of x_{jt} and ε_{jt}, respectively, for all $i \neq j$.

For a finite N, as $T \to \infty$ we have

$$\operatorname*{plim}_{T\to\infty}(\hat{\beta}^a - \beta) = \frac{\mathrm{E}(\bar{y}_{t-1}^2)\,\mathrm{E}(\bar{x}_t\bar{v}_t) - \mathrm{E}(\bar{x}_t\bar{y}_{t-1})\,\mathrm{E}(\bar{y}_{t-1}\bar{v}_t)}{\mathrm{E}(\bar{x}_t^2)\,\mathrm{E}(\bar{y}_{t-1}^2) - \mathrm{E}(\bar{x}_t\bar{y}_{t-1})\,\mathrm{E}(\bar{x}_t\bar{y}_{t-1})}, \tag{B.5}$$

which in view of the above results is in general nonzero. It is also easily seen that when the infinite sums

$$\sum_{s=0}^{\infty} \mathrm{E}(\eta_{2i}\beta_i\lambda_i^s)\,\mathrm{E}(\bar{x}_{jt}x_{i,t-s-1}), \quad \sum_{s=0}^{\infty} \mathrm{E}(\eta_{1i}\beta_i\lambda_i^s)\,\mathrm{E}(\bar{x}_{it}x_{i,t-s-1}), \quad \text{etc.},$$

that appear in the expressions for $\mathrm{E}(\bar{x}_t\bar{v}_t)$ and $\mathrm{E}(\bar{y}_{t-1}\bar{v}_t)$ converge to finite limits, then all the terms in (C.5) are of order N^{-1} and increasing N does not eliminate the inconsistency in the aggregate time-series estimators. We have

$$\lim_{N\to\infty}\operatorname*{plim}_{T\to\infty}(\hat{\beta}^a - \beta) \neq 0,$$

and similarly,

$$\lim_{N\to\infty}\operatorname*{plim}_{T\to\infty}(\hat{\lambda}^a - \lambda) \neq 0.$$

The macro-disturbances, $\bar{v}_t$, are also serially correlated even if the micro-disturbances are white noise processes. In this case assuming for simplicity that the ε_{it}'s are serially and contemporaneously independently distributed, and that x_{it} and x_{jt} are also distributed independently we have

$$\mathrm{E}(\bar{v}_t\bar{v}_{t-s}) = N^{-2}(A + B + C + D + E + F),$$

where

$$A = \sum_{i=1}^{N}\sum_{l=0}^{\infty}\sum_{k=0}^{\infty} \mathrm{E}(\eta_{1i}^2\lambda_i^{l+k})\,\mathrm{E}(\beta_i^2)\,\gamma_i(|s + l - k|)$$

$$+ \sum_{i=1}^{N}\sum_{k=0}^{\infty} \sigma_i^2\,\mathrm{E}(\eta_{1i}^2\lambda_i^{2k+s}),$$

$$B = \sum_{i=1}^{N} \sum_{k=0}^{\infty} \mathrm{E}(\eta_{1i}\eta_{2i}\beta_i\lambda_i^k)\,\gamma_i(|s-k-1|),$$

$$C = \sum_{i=1}^{N} \sum_{k=0}^{\infty} \mathrm{E}(\eta_{1i}\eta_{2i}\beta_i\lambda_i^k)\,\gamma_i(|s+k+1|),$$

$$D = \sum_{i=1}^{N} \mathrm{E}(\eta_{2i}^2)\,\gamma_i(|s|),$$

$$E = \begin{cases} \sum_{i=1}^{N} \sigma_i^2, & s = 0, \\ 0 & s \neq 0, \end{cases}$$

$$F = \begin{cases} 0 & s = 0, \\ \sum_{i=1}^{N} \sigma_i^2\,\mathrm{E}(\eta_{1i}\lambda_i^{s-1}), & s \geqslant 1, \end{cases}$$

where $\sigma_i^2 = \mathrm{E}(\varepsilon_{it}^2)$ and $\gamma_i(|s|) = \mathrm{E}(x_{it}\,x_{i,t-s})$.

Unless one is prepared to make further strong assumptions about the autocovariance function of x_{it}, the above expressions do not simplify further. But it is clear that the serial correlation pattern of $\bar{v}_t$, in the case of micro-relations with lagged dependent variables, are highly complex and depend on higher-order moments of η_{1i} and η_{2i}. Notice that terms such as

$$\mathrm{E}(\eta_{1i}\beta_i\lambda_i^k) = \mathrm{E}\{\eta_{1i}(\lambda + \eta_{1i})^k(\beta + \eta_{2i})\},$$

$$\mathrm{E}(\eta_{1i}\eta_{2i}\beta_i\lambda_i^k) = \mathrm{E}\{\eta_{1i}\eta_{2i}(\beta + \eta_{2i})(\lambda + \eta_{1i})^k\}$$

depend on higher moments of the random variables η_{1i} and η_{2i} and in general do not vanish even when η_{1i} and η_{2i} can be assumed to be distributed independently.

Appendix C

Derivation of the autocorrelation function of time averages

Let $\{x_t\}$ be convariance-stationary process with mean μ and the autocovariance function $\mathrm{E}(x_t - \mu)\,(x_{t-s} - \mu) = \gamma_s = \gamma_{-s}$ and consider time averages $\tilde{x}_{-j} = T^{-}\sum_{t=1}^{T} x_{t-j}$. We are interested in the autocovariance function of the

$\{\tilde{x}_{ij}\}$ process. We have

$$\psi_j = \mathrm{E}[(\tilde{x}_0 - \mu)(\tilde{x}_{-j} - \mu)]$$

$$= \frac{1}{T^2}\sum_{t=1}^{T}\sum_{t'=1}^{T}\gamma_{t-t'+j}$$

$$= \frac{1}{T}\left\{\gamma_j + \sum_{s=1}^{T-1}\left(1 - \frac{s}{T}\right)(\gamma_{j+s} + \gamma_{j-s})\right\}. \tag{C.1}$$

However,

$$\left|\sum_{s=1}^{T-1}\left(1 - \frac{s}{T}\right)\gamma_{j+s}\right| \leqslant \sum_{s=1}^{T-1}|\gamma_{j+s}|,$$

which is bounded since it is assumed that $\{x_t\}$ is covariance-stationary and mean square ergodic. Hence it follows that $\mathrm{E}(\tilde{x}_0\tilde{x}_{-j}) = \mu^2 + \mathrm{O}(T^{-1})$, for all j.

Consider now the term

$$\phi_j = \mathrm{E}\{\tilde{x}_0(\tilde{x}_{-j} - \tilde{x}_{-j-1}\} = \psi_j - \psi_{j+1}.$$

Using (C.1) and after some algebraic manipulations we have

$$\phi_j = \frac{1}{T^2}\left\{\sum_{i=1}^{T}\gamma_{i-j-1} - \sum_{i=1}^{T}\gamma_{i+j}\right\}. \tag{C.2}$$

Once again since $\{x_t\}$ is stationary it follows that the term in $\{\ \}$ is finite and that $\phi_j = \mathrm{O}(T^{-2})$. It is also worth noting that

$$\phi_0 = \frac{1}{T^2}(\gamma_0 - \gamma_T),$$

$$\phi_j = \phi_{j-1} + \frac{1}{T^2}\{(\gamma_j - \gamma_{T+j}) + (\gamma_{j-1} - \gamma_{T-j})\}, \qquad j = 1,2,\ldots.$$

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